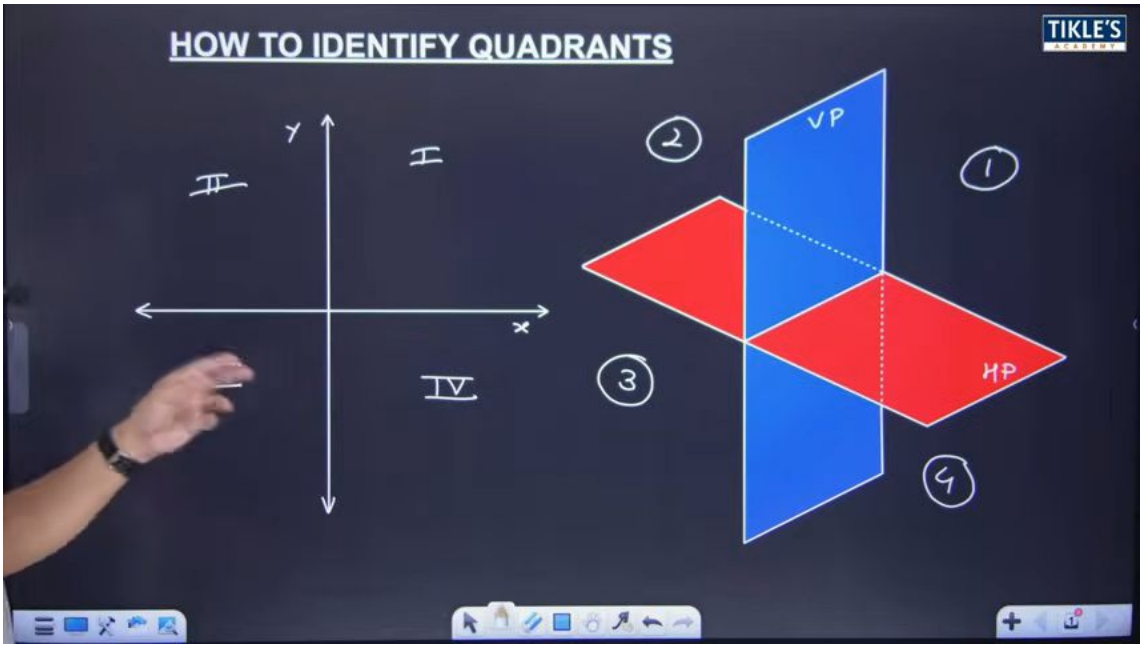
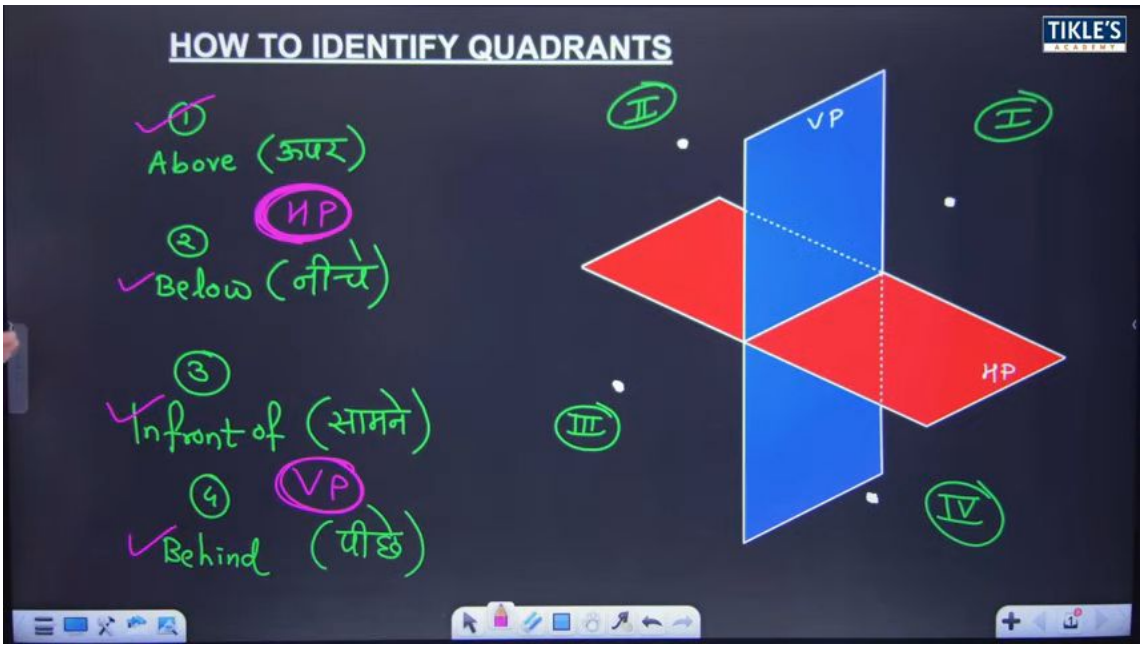
300.141137



667.254504 very imp



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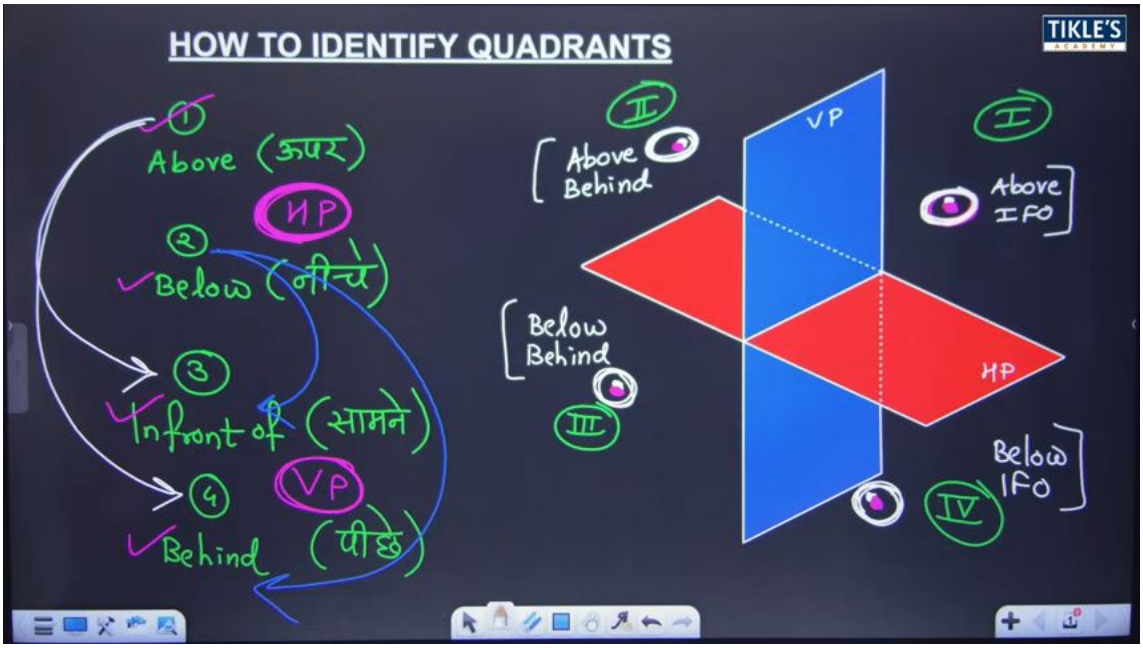
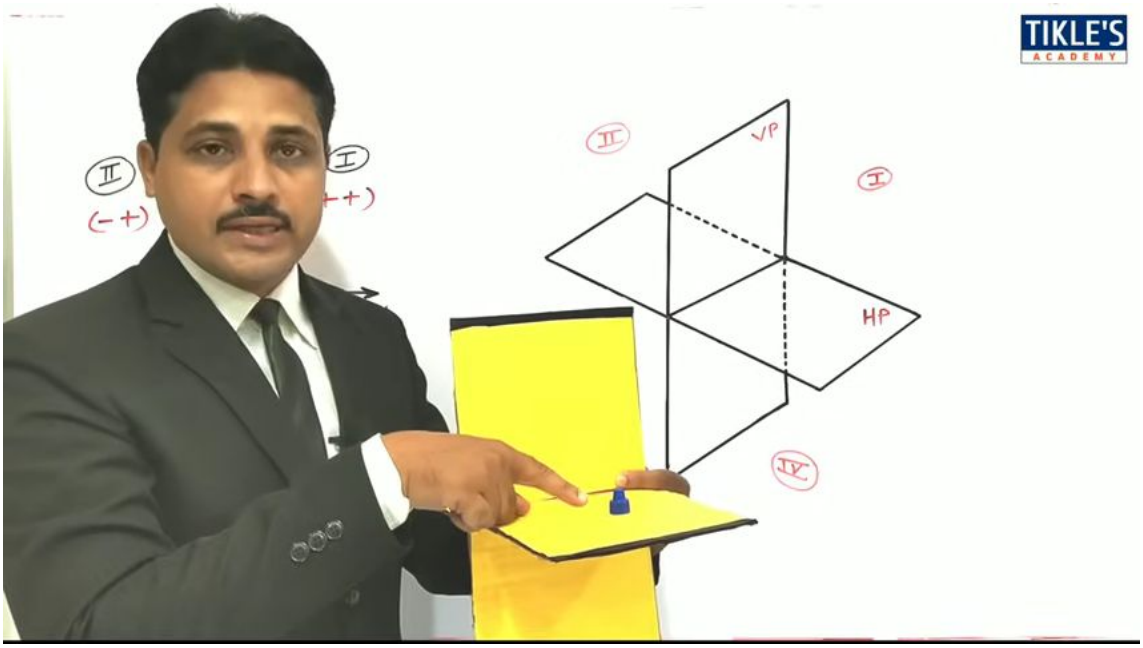
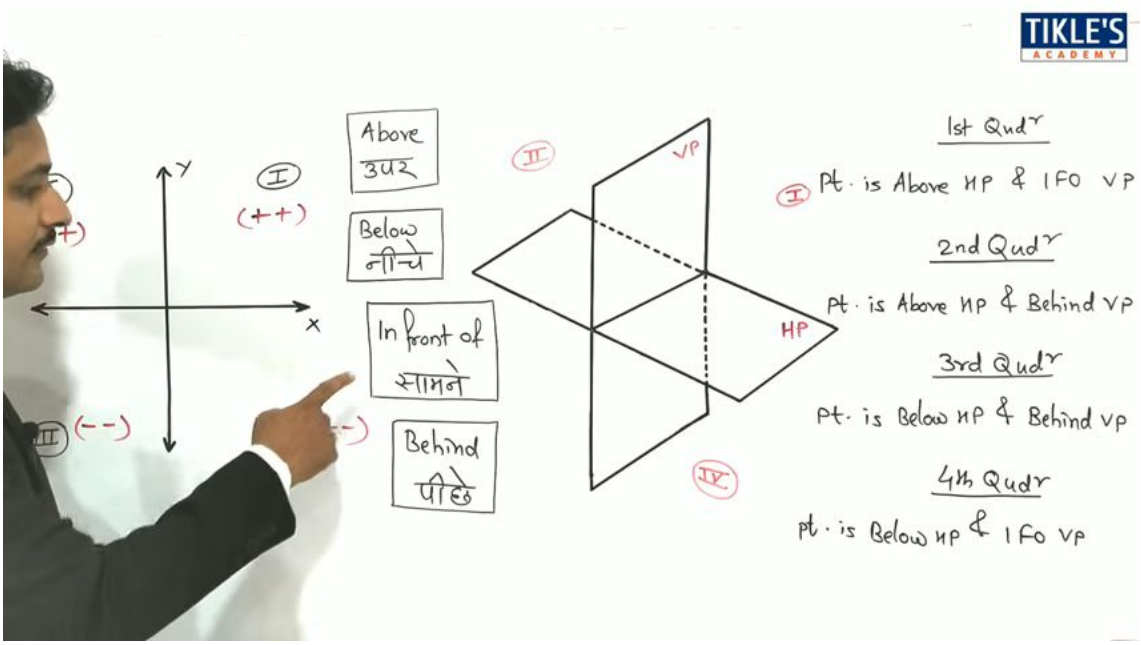


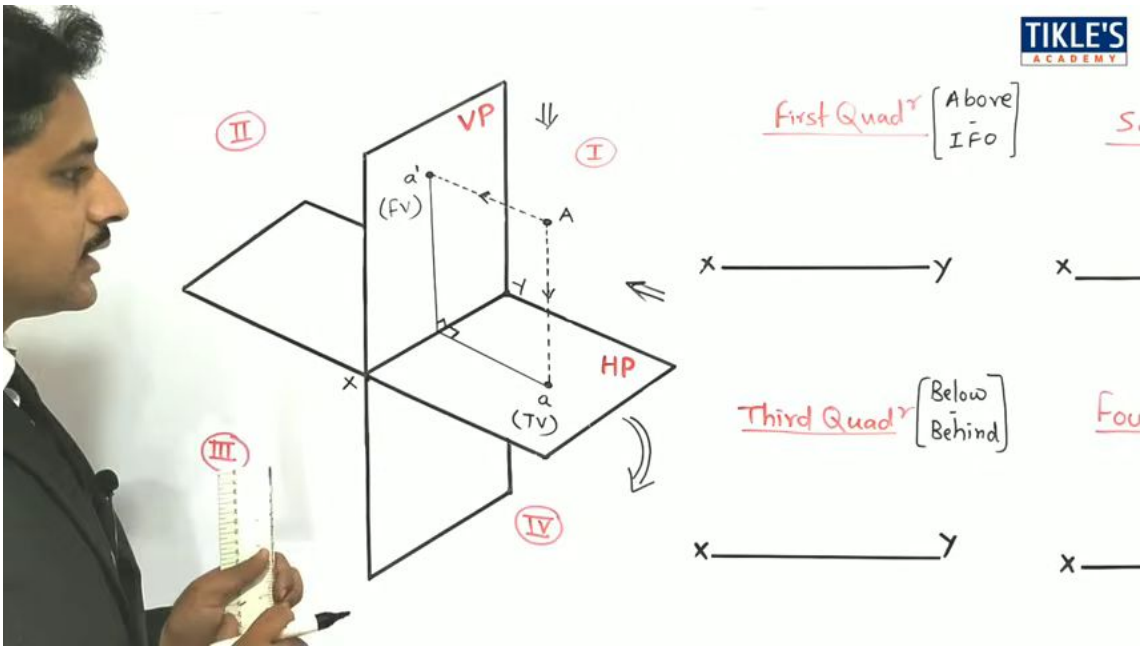
517.926512 Point is in HP



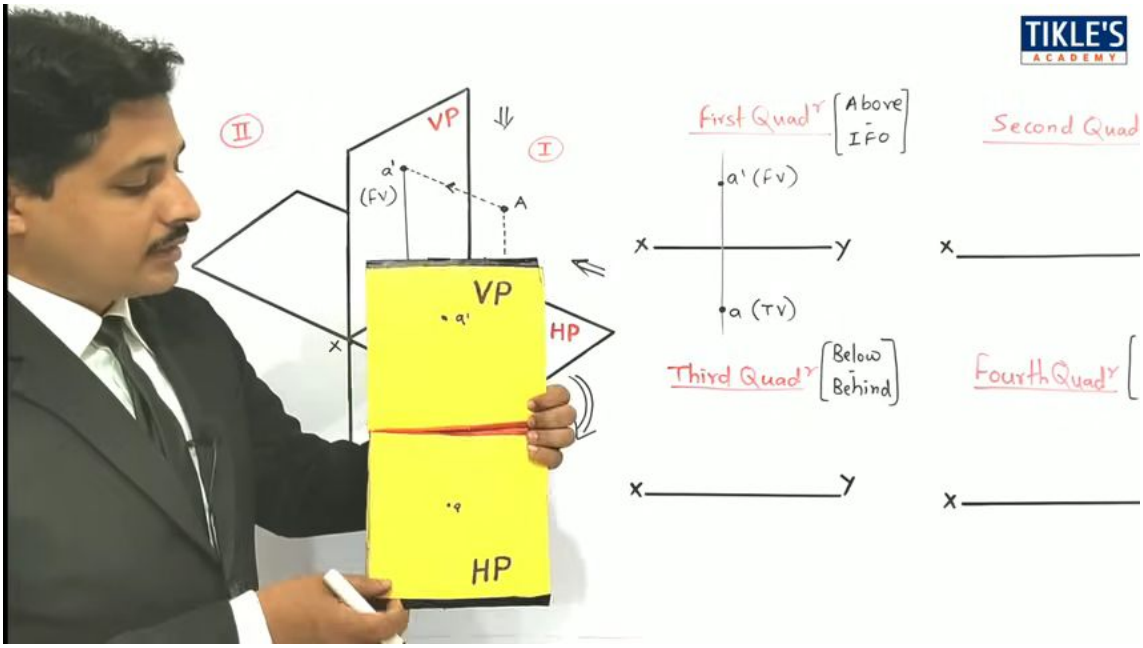
1360.331373



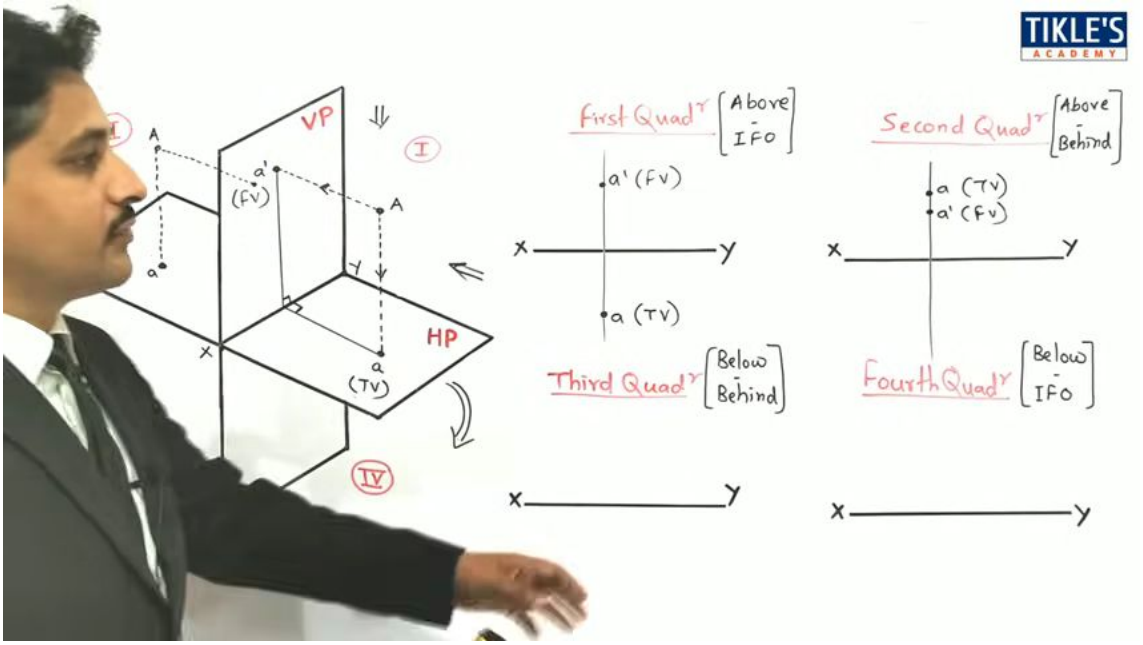
a' - F.V a - T.V 493.830756 first quadrant



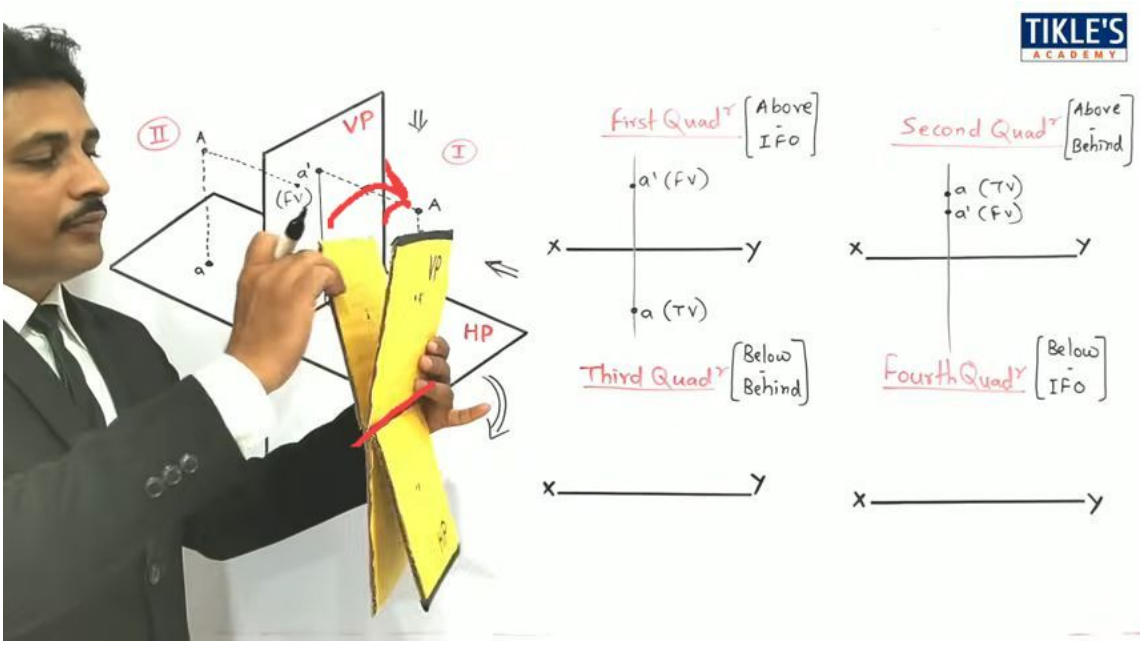
665.845581 if point is in 1st quadrant



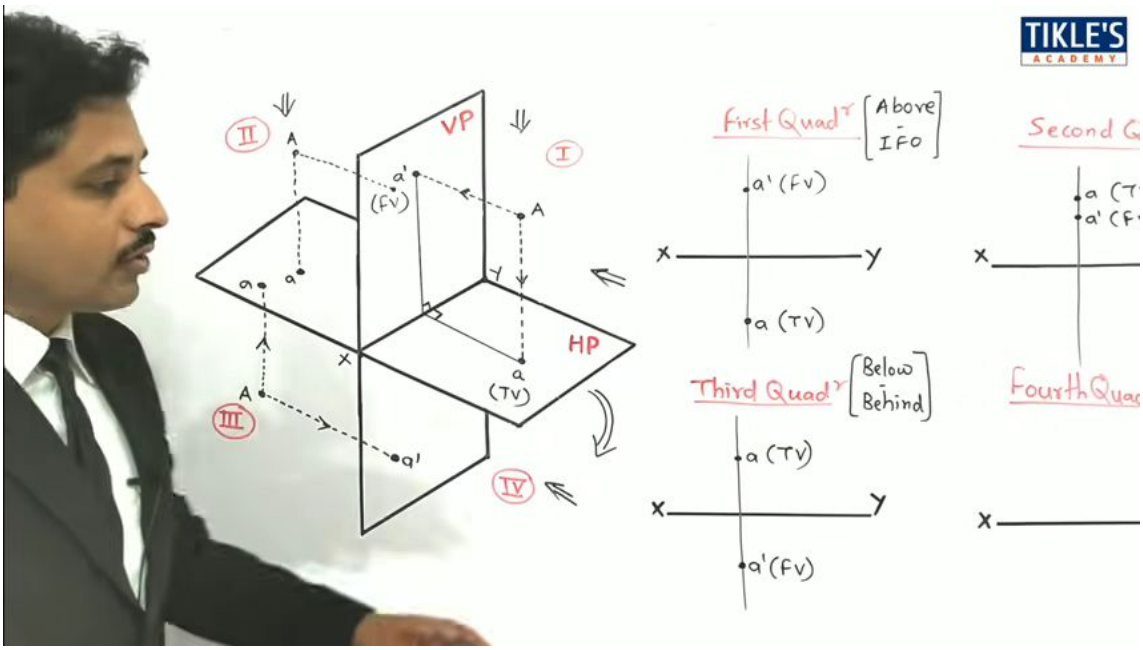
869.446331 second quadrant



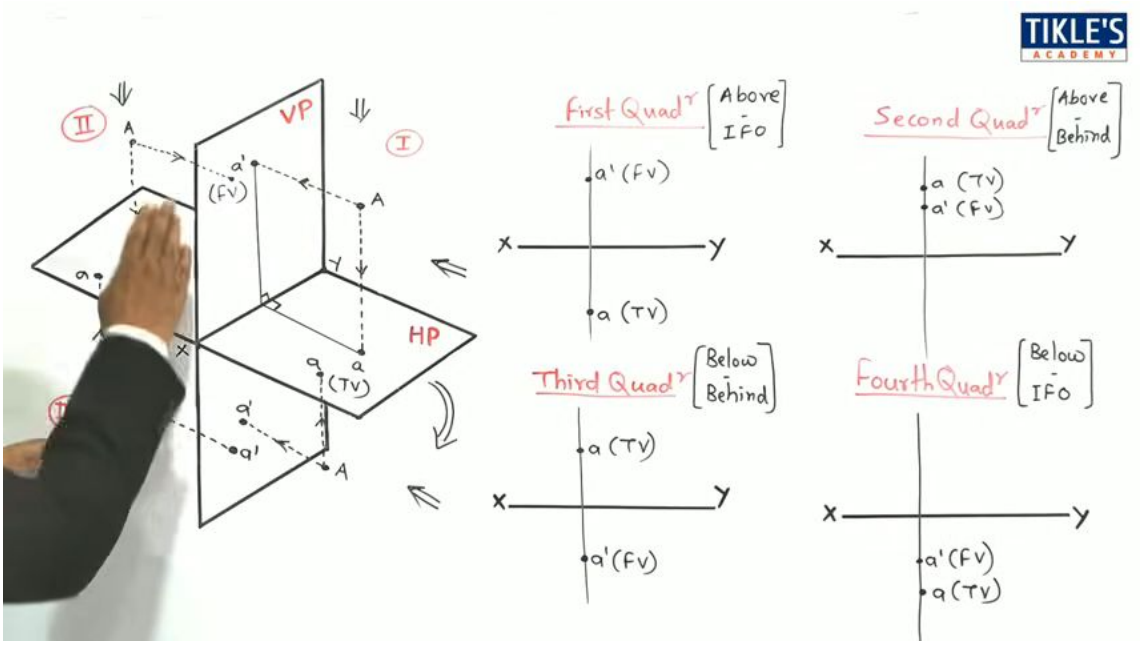
895.359097 since



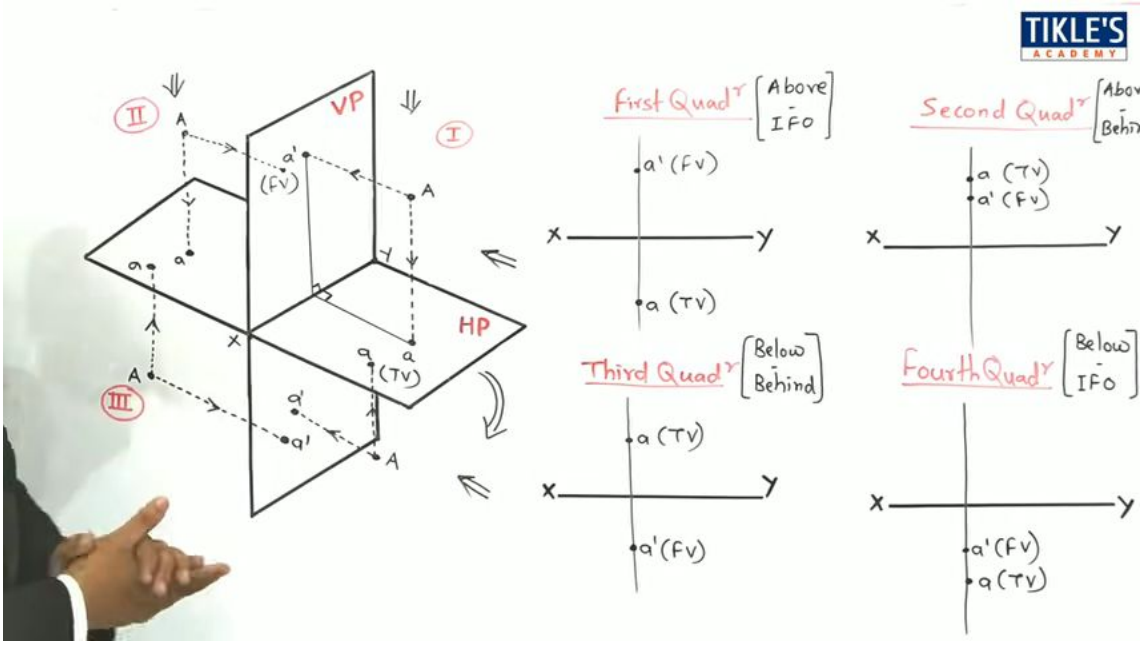
1105.034086 third quadrant



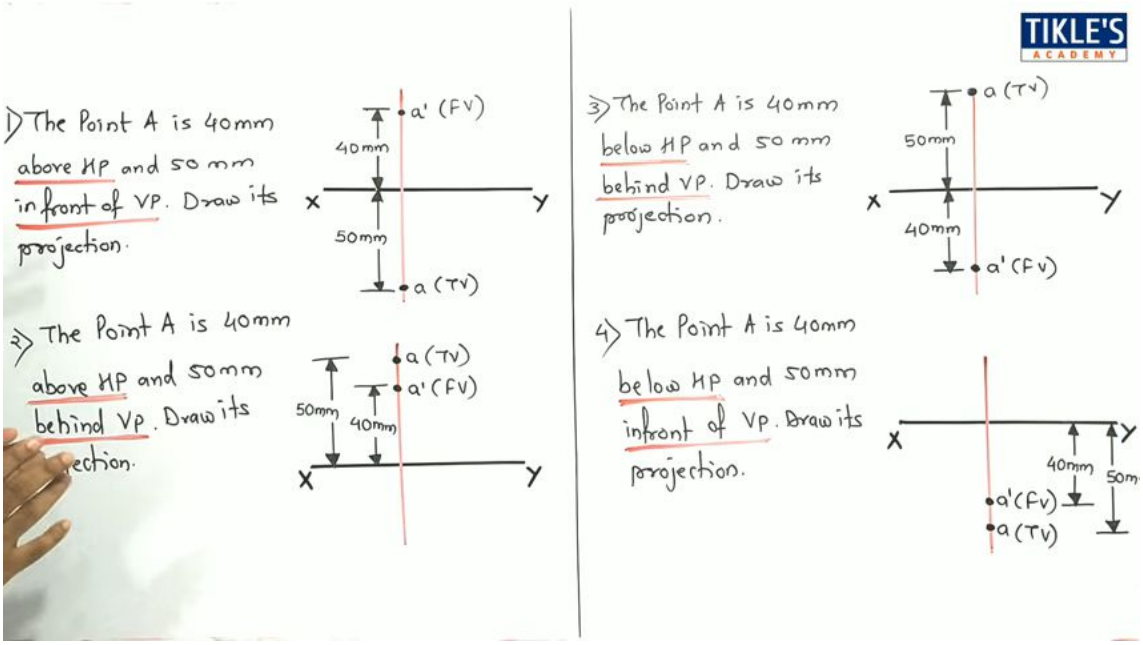
1417.500903 fourth quadrant



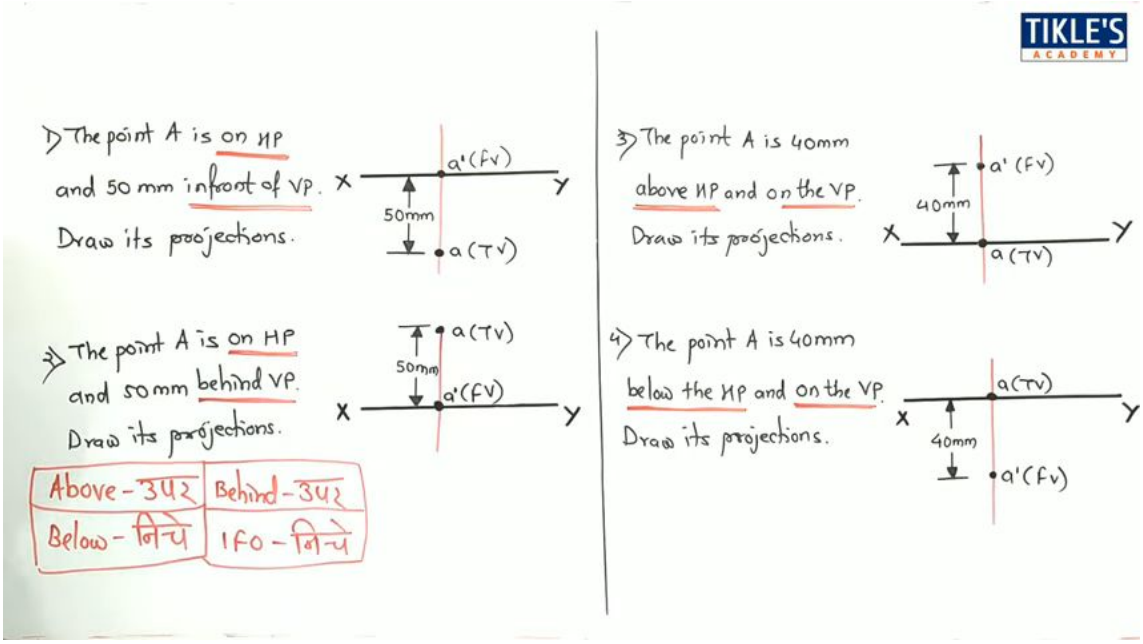
In conclusion, 1421.89428



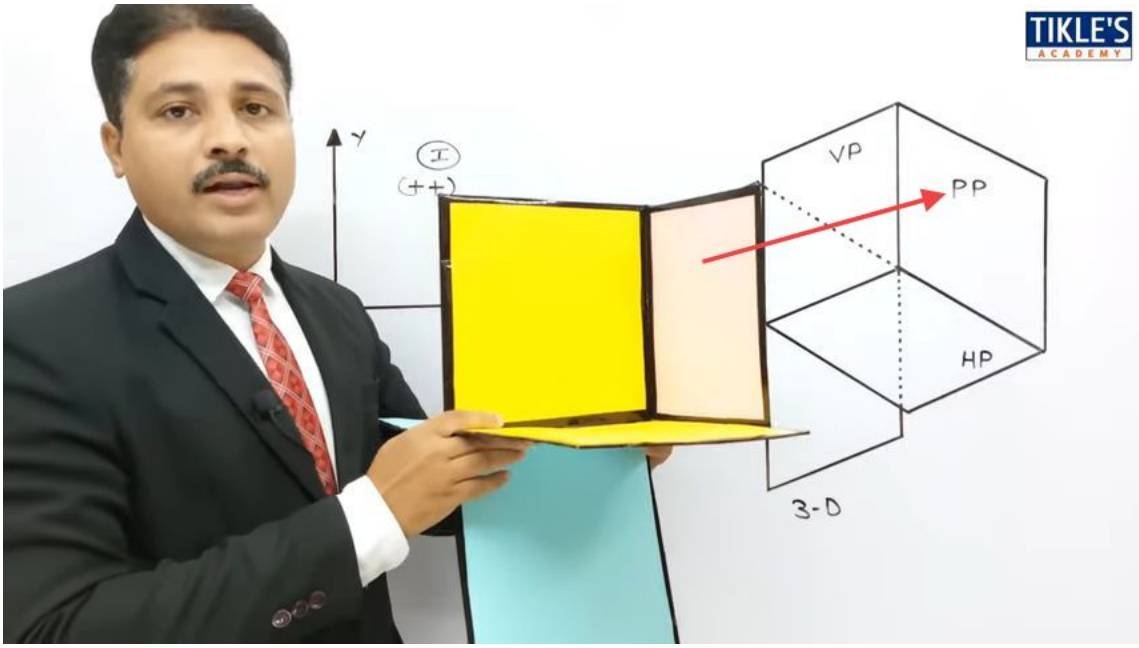
891.245271



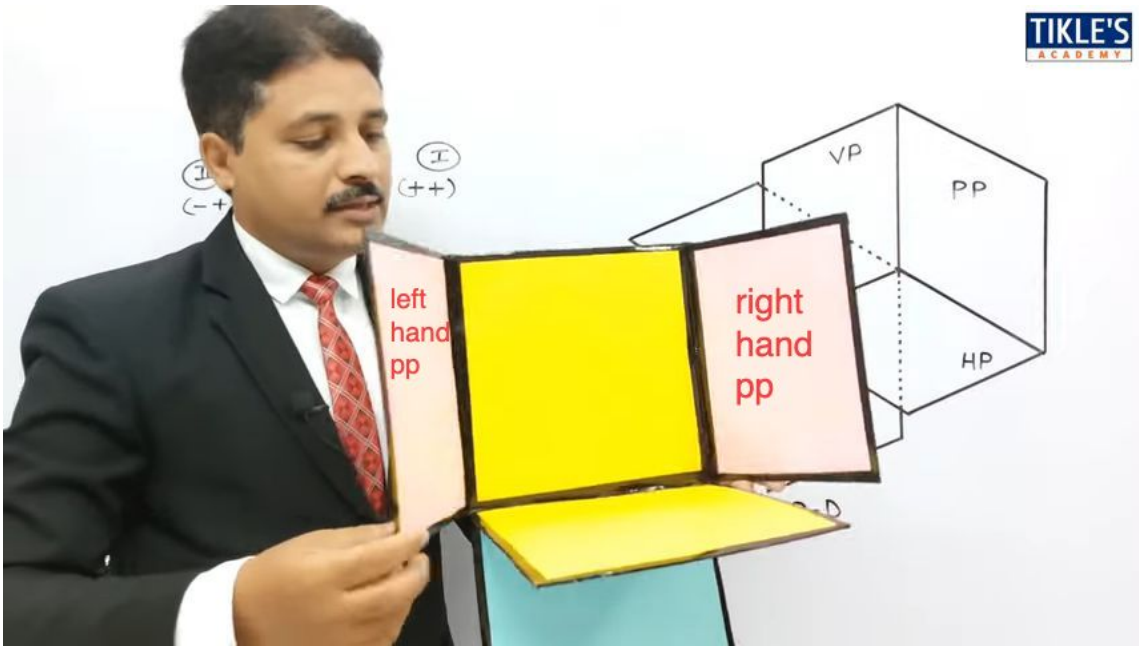
The **HP (Horizontal Plane)** is always **rotated downwards (clockwise)** to align it with the **VP (Vertical Plane)** on the sheet. 948.335154



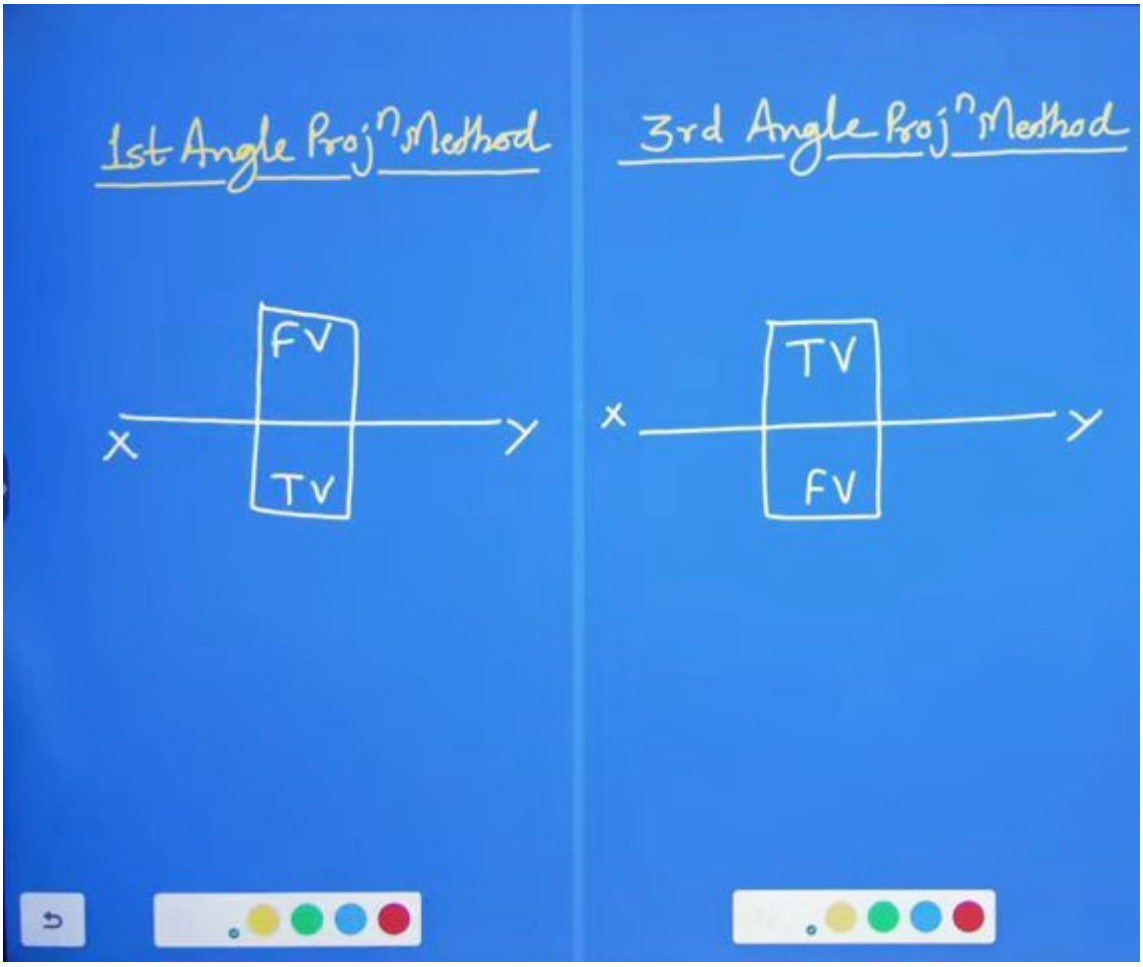
PP - Profile plane. Used to draw side view. It is one of the principle planes/reference planes. 540.15273



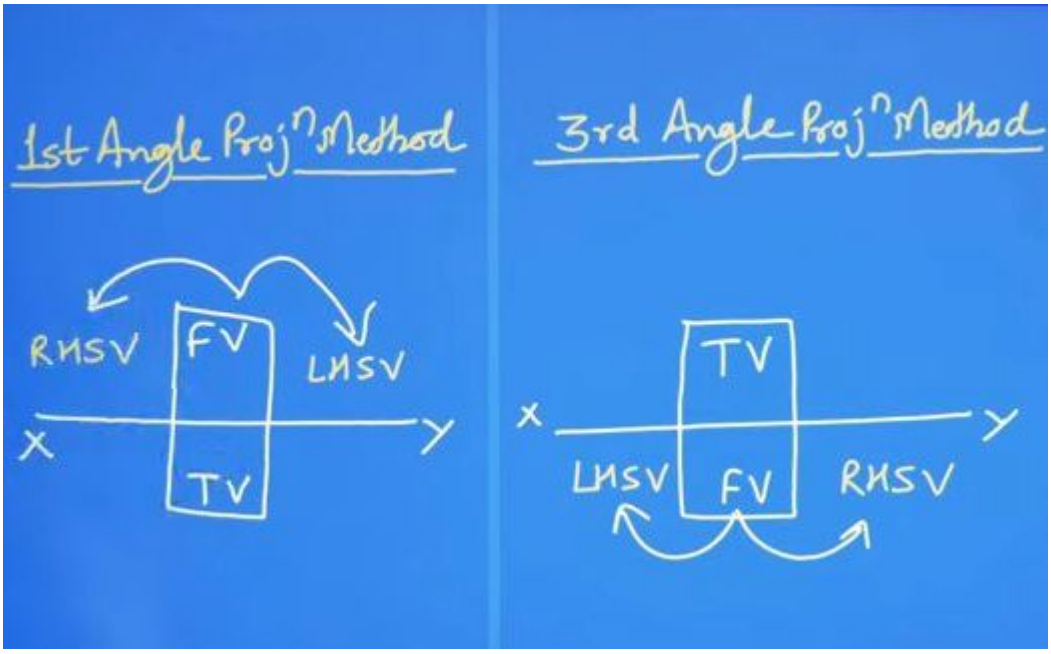
584.360823 there are 2 PPs.



The three planes - VP, HP, PP- are perpendicular to each other. Front View is drawn on VP. Top View is drawn on HP. Side View is drawn on PP. There are 3 reference/principle planes - vp, hp, pp Front View and Top View can be drawn only once. But side view can be drawn twice since two pps. Right-Hand Side View is drawn on the Left-Hand Profile Plane. Left-Hand Side View is drawn on the Right-Hand Profile Plane. XY, where vp and hp intersect, is called the reference line. 311.164318

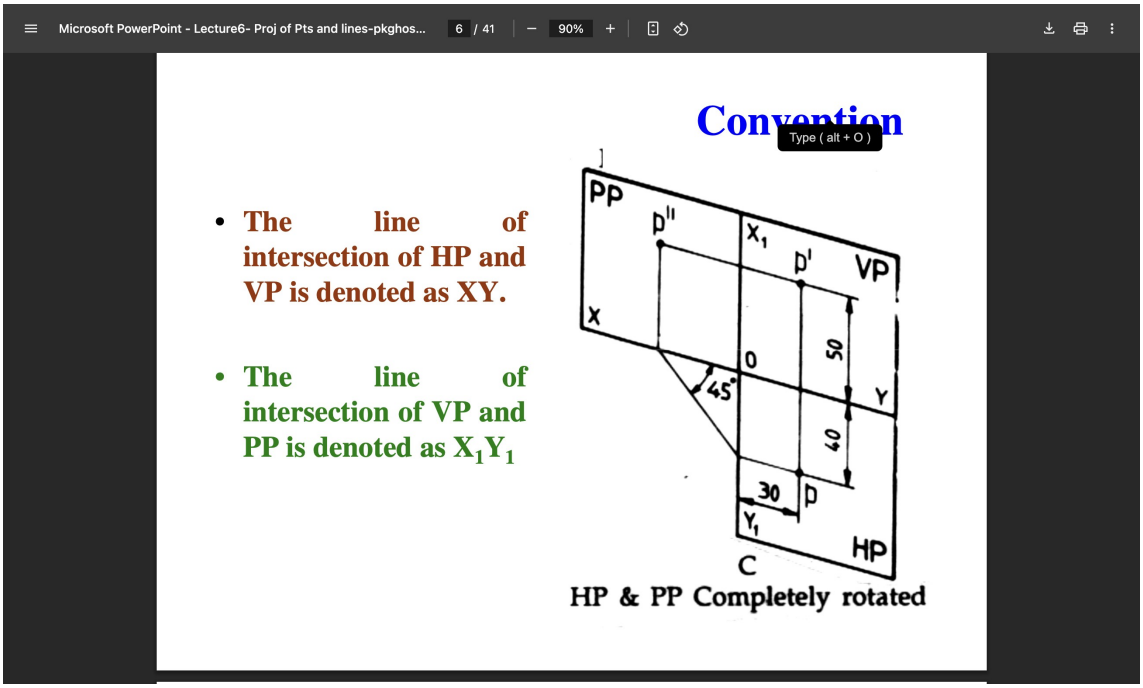


There are two types of side view - LHSV & RHSV. Side views are always on the side of F.V. Never on either side of T.V in either angles projection method. 500.389557



"In front

of LPP" is the RHSV. "In front of RPP" is the LHSV. 0



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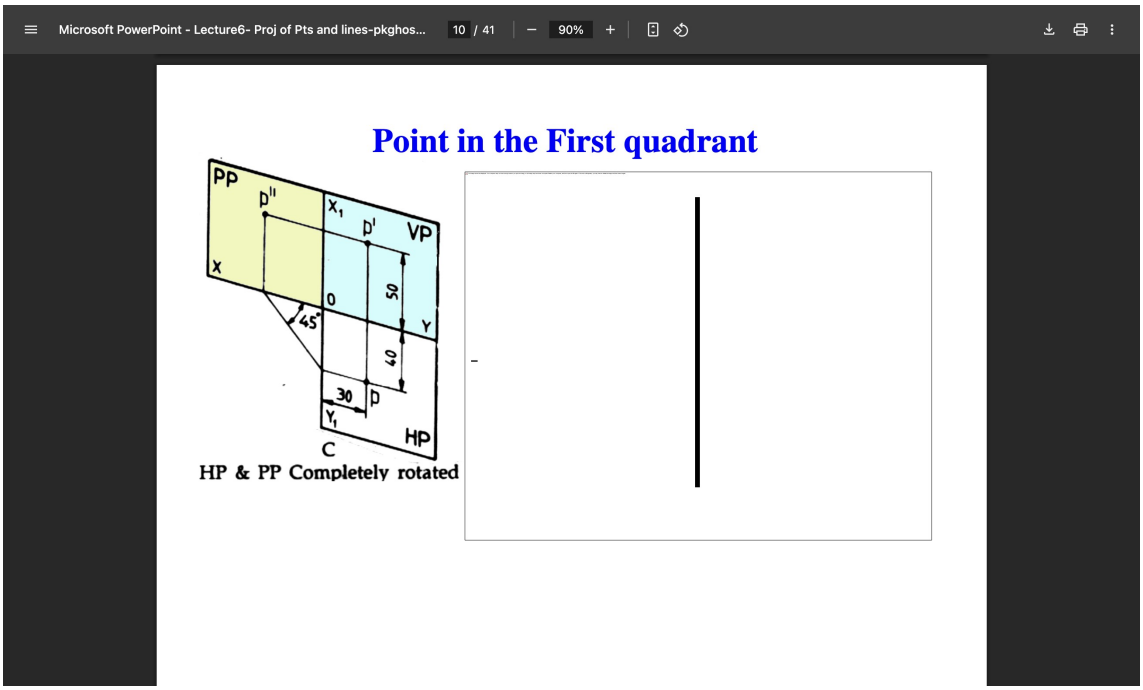
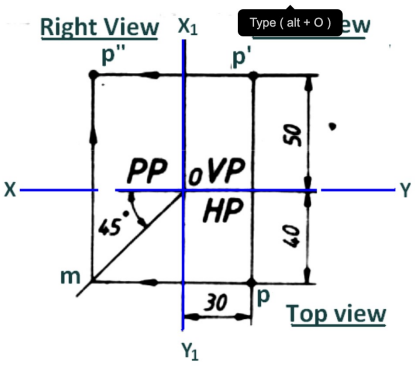


0

Point in the First quadrant

Procedure

- **Draw a thin horizontal line, XY, to represent the line of intersection of HP and VP.**
- **Draw X1Y1 line to represent the line of intersection of VP and PP.**
- **Draw the Top View (p).**
- **Draw the projector line**
- **Draw the Front View (p´)**

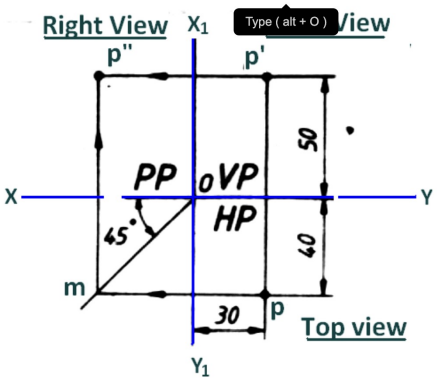


when the procedure says **"Draw the projector line"**, it refers to the **thin vertical black line extending from the top view point p upward towards the front view point p'**, passing through the horizontal line XY. 0

Point in the First quadrant

Procedure

- **To project the right view on the left PP, draw a horizontal projector through p to intersect the 45 degree line at m.**
- **through m draw a vertical projector to intersect the horizontal projector drawn through p´ at p´´.**
- **p´´ is the right view of point P**



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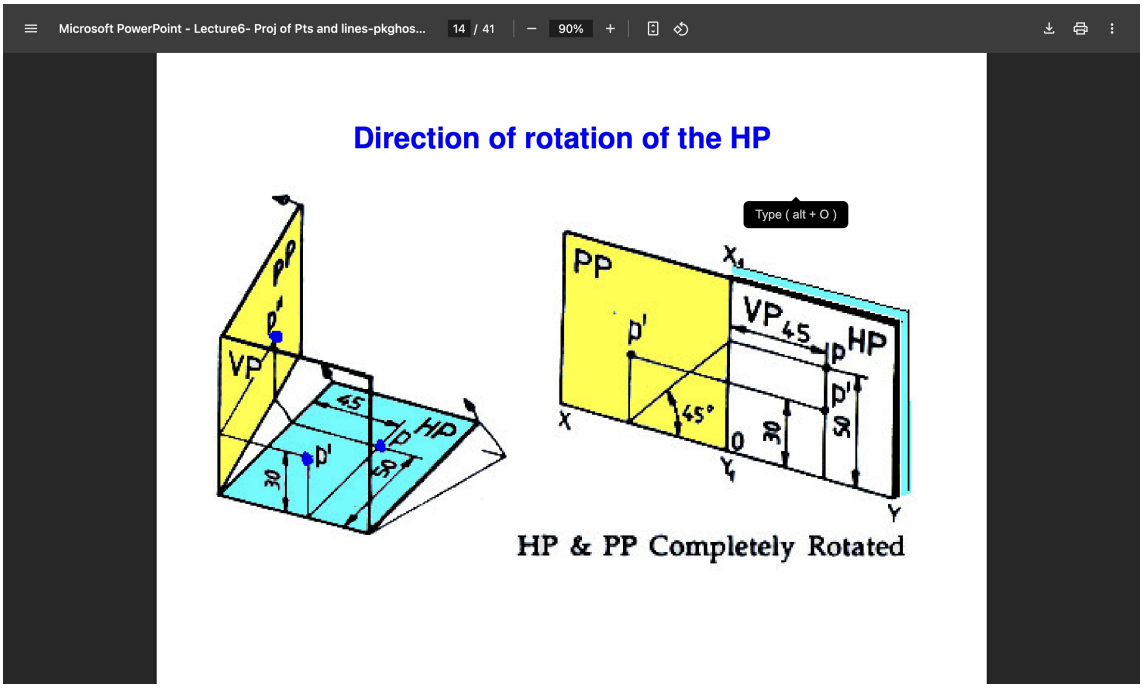
Microsoft PowerPoint - Lecture6- Proj of Pts and lines-pkghos...
14 / 41
90%
Direction of rotation of the HP
Type (alt + O)
PP
VP
HP
p'
p
45
30
50
X
Y
X1
Y1
0
45°
HP & PP Completely Rotated